Ad-hoc Insights

Consumer Goods Domain

Role: Junior Data Analyst | Tools: SQL + Power BI

Presented by Shruthi Bhaskaran M K

Challenge Overview

AtliQ Hardwares

Major computer hardware manufacturer expanding globally across markets.

Critical Gap

Management lacks actionable insights for fast, data-driven decision-making.

Problem Statement

AtliQ Hardwares is experiencing rapid global expansion but struggles to extract actionable business intelligence from existing data systems.

Management Requirements

- Enhanced visibility into sales performance and product metrics
- Real-time financial indicators for strategic planning
- Data-backed decision-making capabilities
- Analysts with strong SQL expertise and business communication skills

Task Requirements

01

Review Business Requests

Analyse 10 ad-hoc requests from management team

02

Write SQL Queries

Develop optimised queries to extract relevant data insights

03

Create Power BI Visuals

Transform SQL results into compelling, actionable visualisations

04

Present to Management

Deliver findings with audio/video presentation (extra weightage)

Business Overview

Global Presence

Operating across multiple continents with expanding market reach in consumer goods sector

Product Portfolio

Comprehensive range of computer hardware, peripherals, and technology solutions

Rapid Growth

Accelerated expansion requiring sophisticated data analytics capabilities

Goals & Objectives

Strategic Goals

Sales Performance

Track revenue trends, regional performance, and customer behaviour patterns

Financial Health

Monitor profitability, cost structures, and fiscal year comparisons

Product Analytics

Identify top performers, underperforming items, and category insights

Deliverable Outcomes

- Actionable insights for executive decision-making
- Clear visualisations for stakeholder communication
- Data-driven recommendations for growth strategies
- Comprehensive analysis spanning FY2019–FY2021

Business Product Hierarchy

1

Division

Top-level categorisation

2

Segment

Market segments within divisions

3

Category

Product categories per segment

4

Product

Individual SKUs and variants

Database Structure Overview

The AtliQ Hardwares database comprises multiple interconnected tables capturing sales transactions, customer information, product details, and financial metrics across fiscal years.

Fact Tables

- Sales Monthly
- Gross Price
- Manufacturing Costs
- Pre Invoice Deductions

Dimension Tables

- Customers
- Products

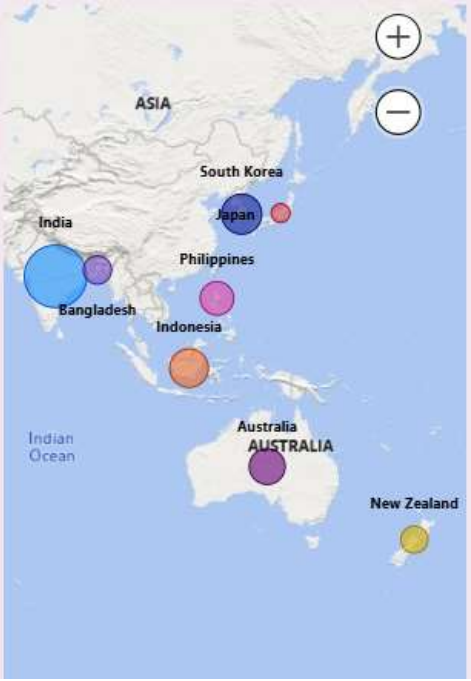
Q1: Markets for "AtliQ Exclusive" in APAC Region

The first ad-hoc request involves identifying all operational markets for our key customer, "AtliQ Exclusive", specifically within the Asia-Pacific (APAC) region. This insight is crucial for strategic resource allocation and targeted business development efforts.

Insights

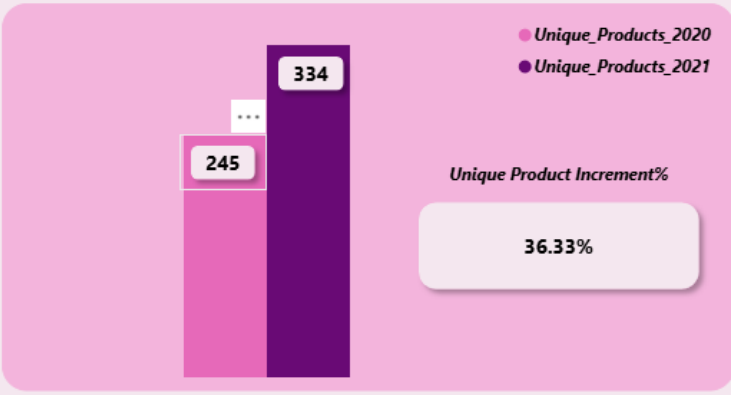
- AtliQ Exclusive operates across multiple APAC markets, indicating a broad distribution presence.
- Presence in these markets strengthens brand recall and supports channel relationships.
- These markets will be essential focus areas when planning year-on-year growth strategies.

Market/Country
Australia
Bangladesh
India
Indonesia
Japan
New Zealand
Philippines
South Korea



Q2: Unique Product Growth Analysis – 2021 vs 2020

Unique_Products_2021	Unique_Products_2020	Percentage_Chg
334	245	36.33



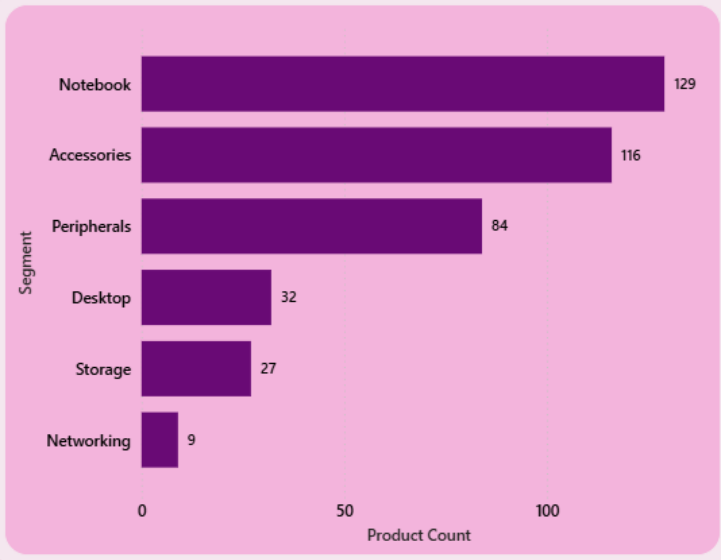
Insights

- Product portfolio expanded from 245 to 334 SKUs between FY2020 and FY2021.
- 36.33% YoY increase in product diversification — a strong expansion signal.
- Suggests investment in capturing more customer segments and responding to market demands.
- Growth in SKUs likely contributed to improved market coverage.

Q3: Unique Product Counts for Each Segment

Insights

- Notebook, Accessories, and Peripherals have high sales volume and are the main revenue drivers.
- Their average sales (~110) indicate strong and consistent customer demand.
- Desktop, Storage, and Networking have very low sales, averaging around 23, indicating declining or niche demand.
- The product portfolio is skewed toward portable and add-on products rather than traditional desktops.



Q4: Most Increased Unique Products Segment-wise – 2021 vs 2020

Insights

- Accessories showed the strongest growth (+34), indicating a major product expansion focus in this segment.
- Notebook and Peripherals saw steady growth (+16 each), supporting core portfolio stability.
- Desktop, though small, recorded a notable relative increase (+15), suggesting targeted market push.
- Storage had minimal change (+5), implying either a mature or lower-priority segment.
- Overall trend shows wider product diversification, led primarily by Accessories.

Segment	Product_Count_2021	Product_Count_2020	Difference
Accessories	103	69	34
Peripherals	75	59	16
Notebook	108	92	16
Desktop	22	7	15
Storage	17	12	5
Networking	9	6	3

Q5: Products by Manufacturing Cost – Highest & Lowest

This analysis identifies products with the most significant manufacturing costs (both highest and lowest) to highlight key cost drivers and potential areas for optimization or strategic pricing.

Understanding these extremes is vital for cost management and profitability analysis. See the accompanying SQL query results and Power BI visuals for the detailed breakdown of highest and lowest manufacturing cost products.

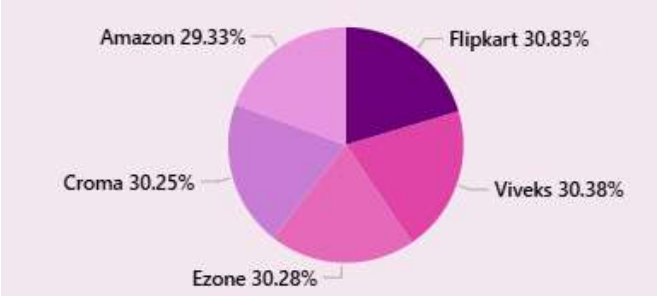
Product_Code	Product	Manufacturing_Cost
A6120110206	AQ HOME Allin1 Gen 2	$240.54
A2118150101	AQ Master wired x1 Ms	$0.89

Q6: Top 5 Customers by Avg Pre-Invoice Discount – FY21 India

Insights

- Flipkart received the highest average pre-invoice discount (30.83%) among Indian market customers in FY2021.
- Viveks (30.38%), Croma (30.28%) and Ezone (30.25%) follow closely — indicating competitive pricing for major retail partners.
- Amazon appears in the top 5 with a slightly lower discount at 29.33%, suggesting tighter margin negotiations.
- Large retail partners received the highest discounts, indicating a volume-driven pricing strategy.

Customer_Code	Customer	AVG_Discount%
90002009	Flipkart	30.83%
90002006	Viveks	30.38%
90002003	Ezone	30.28%
90002002	Croma	30.25%
90002016	Amazon	29.33%



Q7: Monthly Gross Sales Report – AtliQ Exclusive

This analysis provides a complete monthly breakdown of gross sales amounts for the key customer "AtliQ Exclusive". Tracking performance across months and years identifies seasonal patterns, high and low-performing periods, and informs resource allocation and sales planning decisions.

Date	Fiscal Year	Gross Sales Price
Sep 19	2020	4.50
Oct 19	2020	5.14
Nov 19	2020	7.52
Dec 19	2020	4.83
Jan 20	2020	4.74
Feb 20	2020	4.00
Mar 20	2020	0.38
Apr 20	2020	0.40
May 20	2020	0.78
Jun 20	2020	1.70
Jul 20	2020	2.55
Aug 20	2020	2.79
Sep 20	2021	12.35
Oct 20	2021	13.22
Nov 20	2021	20.46
Dec 20	2021	12.94
Jan 21	2021	12.40
Feb 21	2021	10.13
Mar 21	2021	12.14
Apr 21	2021	7.31
May 21	2021	12.15
Jun 21	2021	9.82
Jul 21	2021	12.09
Aug 21	2021	7.18

Q8: Quarter with Maximum Total Sold Quantity – FY2020

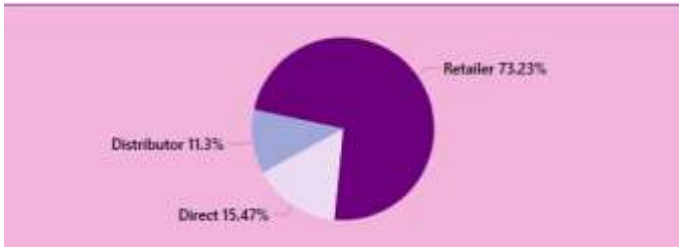
Key Insights

- Q1 is the top-performing quarter with 7.01M units — driven mainly by November and October.
- Q2 remains strong at 6.65M, showing stable performance despite seasonal fluctuations.
- Q4 holds third place with 5.04M, boosted slightly by June.
- Q3 is the weakest quarter at 2.08M, reflecting low sales especially in March and April.
- Overall, the business experiences a clear early-year peak (Q1–Q2) and a significant dip in mid-year (Q3).

Quarter	Total Sold Quantity
Q1	7.01M
Q2	6.65M
Q4	5.04M
Q3	2.08M

Q9: Channel Gross Sales Contribution – FY21

Channel	Gross_Sales_Mln	Percentage_Contribution
Retailer	*1,219.08*	*73.23%*
Direct	*257.53*	*15.47%*
Distributor	*188.03*	*11.30%*



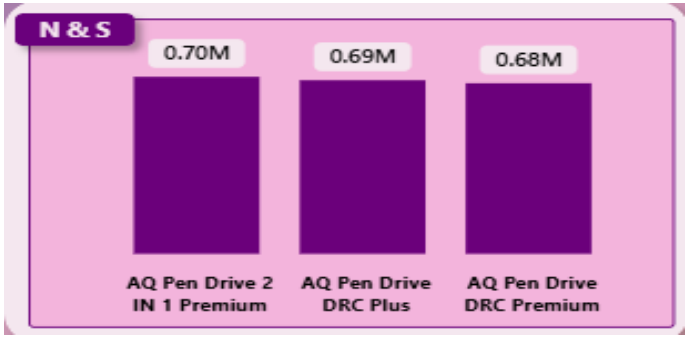
Insights

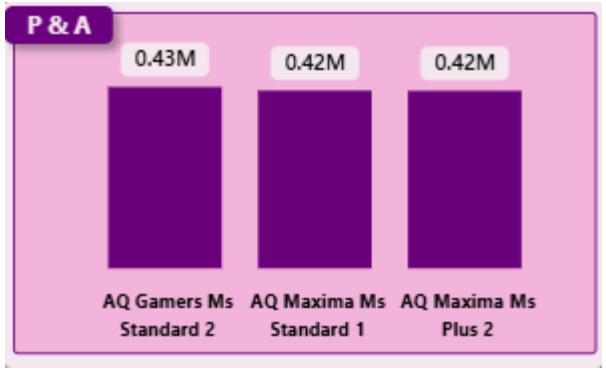
- Retail is the dominant channel (~73%), the largest and strongest revenue driver.
- Direct contributes ~15%, showing steady but secondary performance.
- Distributor at ~11% indicates opportunity for expansion.

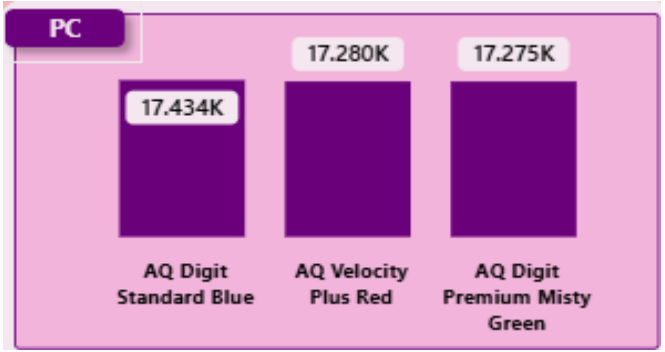
Q10: Top 3 Products by Division – FY21 Total Sold Quantity

This analysis identifies the top 3 products in each division based on total sold quantity during FY2021. Understanding top performers within specific divisions helps in strategic planning, inventory management, and marketing for high-demand items.

division	product_code	product_variant	total_qty
N & S	A6720160103	AQ Pen Drive 2 IN 1 Premium	701373
N & S	A6818160202	AQ Pen Drive DRC Plus	688003
N & S	A6819160203	AQ Pen Drive DRC Premium	676245
P & A	A2319150302	AQ Gamers Ms Standard 2	428498
P & A	A2520150501	AQ Maxima Ms Standard 1	419865
P & A	A2520150504	AQ Maxima Ms Plus 2	419471
PC	A4218110202	AQ Digit Standard Blue	17434
PC	A4218110208	AQ Digit Premium Misty Green	17275
PC	A4319110306	AQ Velocity Plus Red	17280







Thank You

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